Exercise

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Game Theory

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Problem 1

We have an economy with the following characteristics:

- Two classes: Workers and Capitalists.
- Both classes have a constant propensity to save.
- Labor and capital markets are perfectly competitive.
- Workers' income sources are wages and profits, while capitalists' income source is only profits.
- The economy operates in discrete time.
- There is a single good produced.
- Workers save differently from labor and capital income.
- The production function uses CES with the form of:

$$f(k) = (\alpha + (1-\alpha)k^{\rho})^{\frac{1}{\rho}}$$

Where:

- k represents the total capital (sum of capital owned by workers, k_w, and capitalists, k_c).
- s_{ww} represents workers' saving out of wages.
- s_{wp} represents workers' saving out of capital's revenue.
- s_c represents capitalists' savings.
- We assume $s_c > \max s_{ww}, s_{wp}$.

(a)

$$G(k_w, k_c) = \left(\frac{s_{ww}f(k) - kf'(k)}{s+\delta}, \frac{s_c f'(k)k_c}{s+\delta}\right)$$